# 50 FX Derivatives Problems for Quant Interviews

Practice Workbook with Complete Solutions

For pricing, strats, model validation, market risk, and quantitative research interviews

These are original, representative interview-style problems built around common FX-derivatives themes. They are **not** claimed to be verbatim questions from any specific bank or employer.

Includes forwards, swaps, NDFs, Garman–Kohlhagen, Greeks, smile, barriers, quanto, collateral/basis, calibration, P&L explain, Monte Carlo, PDEs, and model validation.

# Contents

# 1 Formula Reference

Throughout, $S_t$ denotes domestic currency units per one unit of foreign currency, $r_d$ and $r_f$ are domestic and foreign continuously compounded rates, $T$ is maturity, and $\sigma$ is annualized volatility.

### Covered interest parity

$$F(0,T) = S_0 e^{(r_d - r_f)T} = S_0 \frac{P_f(0,T)}{P_d(0,T)}.$$

### Garman–Kohlhagen

$$C = S_0 e^{-r_f T} N(d_1) - K e^{-r_d T} N(d_2), \qquad P = K e^{-r_d T} N(-d_2) - S_0 e^{-r_f T} N(-d_1),$$

$$d_1 = \frac{\ln(S_0/K) + (r_d - r_f + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}.$$

### FX put–call parity

$$C - P = S_0 e^{-r_f T} - K e^{-r_d T}.$$

### Core Greeks

$$\Delta^C_{\text{spot}} = e^{-r_f T} N(d_1), \qquad \Delta^C_{\text{fwd}} = N(d_1),$$

$$\Gamma = \frac{e^{-r_f T} n(d_1)}{S_0 \sigma \sqrt{T}}, \qquad \nu = S_0 e^{-r_f T} n(d_1)\sqrt{T}.$$

### FX smile quotes

$$RR_{25} = \sigma_{25C} - \sigma_{25P}, \qquad BF_{25} = \frac{\sigma_{25C} + \sigma_{25P}}{2} - \sigma_{\text{ATM}}.$$

### Risk-neutral FX dynamics

$$dS_t = (r_d - r_f) S_t \, dt + \sigma S_t \, dW_t.$$

### Quanto intuition

The quanto drift correction is proportional to $\rho\sigma_X\sigma_S$, with sign determined by the precise FX quote, numeraire, and payoff convention.

## 2 Problems

### Problem 1
[Notation] – Easy

EUR/USD is quoted at 1.0875. Under the convention S = domestic currency per unit of foreign currency, identify the domestic and foreign currencies. If EUR/USD rises to 1.0950, who benefits: a trader long EUR/USD or short EUR/USD?

### Problem 2
[Notation] – Easy

USD/JPY moves from 149.20 to 149.68. How many pips is the move? If the platform shows 149.683, what is the final decimal digit commonly called?

### Problem 3
[Notation] – Easy

A corporate treasurer expects to receive EUR 5 million in three months and reports in USD. Explain the FX exposure and the sign of the hedge using EUR/USD forwards.

### Problem 4
[Arbitrage] – Medium

Suppose EUR/USD = 1.1000, USD/JPY = 150.00, while EUR/JPY is quoted at 166.50. Is there triangular arbitrage? Construct one cycle starting with JPY 150,000,000 and compute the approximate profit ignoring bid-ask spreads.

### Problem 5
[Quote Inversion] – Medium

If EUR/USD = 1.2500, compute USD/EUR. If EUR/USD increases by 1%, approximately what percentage change occurs in USD/EUR? Explain why the sign flips.

### Problem 6
[Forwards] – Medium

Derive continuous-compounding covered interest parity (CIP): F(0,T) = S0 exp((r_d-r_f)T), using a no-arbitrage cash-and-carry argument.

### Problem 7
[Forwards] – Medium

EUR/USD spot is 1.0800, USD rate 4.5%, EUR rate 2.5%, T = 0.5 years, continuously compounded. Compute the 6M forward outright and forward points.

### Problem 8
[Forwards] – Medium

Under annual simple compounding, derive the forward formula F = S(1+r_d T)/(1+r_f T). Using S=1.3000, r_d=5%, r_f=3%, T=0.25, compute F.

### Problem 9
[Forwards] – Medium

A dealer quotes 3M EUR/USD forward points +28 and 6M points +61, where one pip is 0.0001. Estimate 4M points by linear interpolation in time and state one limitation of this approach.

### Problem 10
[FX Swap] – Medium

Spot EUR/USD is 1.0900 and the 3M forward is 1.0945. A client sells EUR 10 million spot and buys EUR 10 million back forward. Describe the two legs and compute the USD cash flows, ignoring settlement timing details.

### Problem 11
[NDF] – Medium

Explain the mechanics of a USD/INR NDF when the contract notional is USD 1 million, NDF rate K = 84.00 INR/USD, and fixing is 85.20. Assume USD cash settlement using payoff USD = N* (Fix-K)/Fix for a party long USD.

### Problem 12
[NDF] – Medium

For the same NDF convention, N = USD 2 million, K = 83.50, fixing = 82.80. Compute the USD settlement to the long-USD party and interpret the sign.

### Problem 13
[NDF Risk] – Hard

An INR NDF fixes tomorrow. Explain fixing risk, basis risk between onshore and offshore markets, and why hedging with spot or futures may not eliminate settlement P&L.

### Problem 14
[GK Derivation] – Hard

Starting from risk-neutral FX dynamics dS=(r_d-r_f)S dt + sigma S dW under the domestic measure, derive the Garman-Kohlhagen European call formula and define d1 and d2.

### Problem 15
[GK Numerical] – Medium

Price a 6M EUR/USD call with S=1.1000, K=1.1200, r_d=4%, r_f=2%, sigma=12%, T=0.5. Use Garman-Kohlhagen. Give value in USD per EUR notional.

### Problem 16
[Put-Call Parity] – Medium

Derive FX put-call parity C-P = S exp(-r_f T) - K exp(-r_d T). For S=1.20, K=1.18, r_d=3%, r_f=1%, T=1, and C=0.075, compute P.

### Problem 17
[Synthetic] – Medium

Show how a long FX call plus a short FX put with the same strike and maturity replicates a forward-like payoff. Explain the discounted cash-flow adjustment relative to a zero-cost forward.

### Problem 18
[Delta] – Medium

For an unadjusted call with d1=0.35, r_f=2%, T=0.5, compute spot delta and forward delta. Use N(0.35)=0.6368.

## Problem 19

[Delta Conventions] – Hard

Explain spot delta, forward delta, and premium-adjusted delta in FX options. Why can a 25-delta call have different strikes under different conventions?

## Problem 20

[Greeks Numerical] – Medium

For S=1.10, r_f=2%, sigma=10%, T=0.25, d1=0.20, and n(d1)=0.3910, compute approximate gamma and vega under Garman-Kohlhagen.

## Problem 21

[ATM] – Medium

Explain ATM-spot, ATM-forward, and delta-neutral ATM conventions. Which strike equals F under ATM-forward? Why is ATM not universally S in FX?

## Problem 22

[Smile] – Medium

Market quotes: ATM vol = 10.0%, 25D RR = -1.2 vol points, 25D BF = 0.5 vol points. Using RR = sigma_25C - sigma_25P and BF=(sigma_25C+sigma_25P)/2 - sigma_ATM, recover 25D call and put vols.

## Problem 23

[Strike From Delta] – Hard

Assume unadjusted forward delta for a call: Delta_F=N(d1)=0.25. For F=1.1000, sigma=12%, T= 0.5, derive K from d1 and compute it using N^{-1}(0.25)=-0.67449.

## Problem 24

[Smile Interpolation] – Hard

You have implied vols at 10D put, 25D put, ATM, 25D call, 10D call for two maturities. Propose a robust interpolation scheme across strike/delta and time. Identify at least three static-arbitrage checks.

## Problem 25

[Static Arbitrage] – Hard

State and explain the no-arbitrage shape restrictions on European call prices as functions of strike and maturity. Relate them to monotonicity, convexity, and calendar arbitrage.

## Problem 26

[Local Vol] – Hard

Explain the intuition of Dupire local volatility. Why can a local-vol model fit the full vanilla surface exactly yet still misprice barrier options?

## Problem 27

[Heston FX] – Hard

Write Heston-style FX dynamics under the domestic measure, including stochastic variance. Explain the roles of r_d, r_f, kappa, theta, xi, and rho.

## Problem 28

[Correlation] – Hard

In a stochastic-vol FX model, what qualitative effect does negative spot-vol correlation have on the implied-volatility skew? Explain for a quote S = domestic per foreign.

## Problem 29

[SABR] – Hard

Describe how SABR-style parameters alpha, beta, rho, nu affect an FX smile. Why can calibration become unstable if all parameters are free at every maturity?

## Problem 30

[Calibration] – Hard

Design a calibration objective for a stochastic-vol FX model using ATM, 25D RR/BF, and 10D RR/BF across maturities. Discuss weighting by bid-ask, regularization, and parameter stability.

## Problem 31

[Barrier] – Medium

Define up-and-out, up-and-in, down-and-out, and down-and-in FX options. For a vanilla call value of 4.2 cents and matching knock-out value of 2.7 cents, use in-out parity to infer the knock-in value.

## Problem 32

[Barrier Monitoring] – Medium

A continuously monitored up-and-out call has barrier H=1.1500. A discretely monitored version checks only daily. Which is generally more valuable, all else equal, and why?

## Problem 33

[One-Touch] – Hard

A one-touch pays USD 1 million immediately if `EUR/USD` ever reaches 1.1500 before 3M expiry. Explain the key risk drivers and why its hedge near the barrier can be unstable.

## Problem 34

[Double Barrier/TARF] – Hard

Explain conceptually how a double-no-touch differs from a TARF. Identify the principal model risks of each.

## Problem 35

[Asian] – Medium

An Asian FX call is based on the arithmetic average of four fixings: 1.0900, 1.1000, 1.1200, 1.1100, strike 1.1050. Compute the payoff per EUR notional at maturity.

## Problem 36

[Quanto] – Hard

A foreign equity follows dX/X = mu_X dt + sigma_X dW_X, and FX follows dS/S = ... + sigma_S dW_S with corr(dW_X,dW_S)=rho. Explain why a quanto payoff fixed in domestic currency acquires a correlation-dependent drift adjustment. State the common sign form under a standard domestic-measure setup.

## Problem 37

[Triangular Consistency] – Medium

Given EUR/USD=1.0800 and GBP/USD=1.2600, compute the consistent EUR/GBP cross rate. If the market EUR/GBP is 0.8500, is it approximately consistent?

## Problem 38

[Cross-Gamma] – Hard

A portfolio is long EUR/USD options and GBP/USD options. Explain why a simultaneous USD shock can create cross-gamma/correlation risk not captured by summing standalone gammas.

## Problem 39

[Collateral] – Hard

Explain why the collateral currency in a CSA can alter the valuation of an FX forward or option. Relate the argument to discounting curves and collateralized no-arbitrage.

## Problem 40

[OIS Forward] – Hard

Spot EUR/USD = 1.1000. USD OIS discount factor P_d(0,T)=0.9700 and EUR OIS discount factor P_f(0,T)=0.9850 for the same maturity. Compute the collateral-consistent forward F=S*P_f/P_d.

## Problem 41

[Cross-Currency Basis] – Hard

Explain cross-currency basis as a deviation from textbook CIP once funding and collateral conventions are included. What market instruments reveal the basis?

## Problem 42

[CSA Numerical] – Medium

Two valuation setups imply 1Y discount factors P_USD=0.955 and P_EUR=0.975. For EUR/USD spot 1.0800, compute the implied forward. If an alternative collateral setup gives P_USD=0.960 and P_EUR=0.972, compute the new forward and difference in pips.

## Problem 43

[P&L Explain] – Hard

An FX option book has delta +EUR 4m, gamma +EUR 0.8m per (1% spot)^2, vega +USD 25k per vol point, and theta -USD 8k/day. Spot rises 0.5%, implied vol rises 0.4 vol points over one day. Estimate first/second-order P&L, treating delta P&L in USD using spot 1.10.

## Problem 44

[Discrete Hedging] – Hard

Explain the gamma-theta trade-off for a delta-hedged long vanilla option. Why does realized volatility above implied volatility tend to help a long-gamma position under idealized assumptions?

## Problem 45

[Vanna/Volga] – Hard

Define vanna and volga. Explain why a smile-sensitive exotic can have small Black-Scholes vega but large vanna/volga risk.

## Problem 46

[Vanna-Volga] – Hard

You have three liquid benchmark vanillas: 25D put, ATM, 25D call. Explain the vanna-volga method conceptually as matching vega, vanna, and volga of an exotic with a portfolio of benchmarks, then pricing a smile correction.

## Problem 47

[Monte Carlo] – Hard

Design a Monte Carlo pricer for a 1Y `EUR/USD` Asian call under lognormal GK dynamics. Specify simulation equation, discounting, estimator, standard error, and two variance-reduction methods.

## Problem 48

[PDE/Numerics] – Hard

Write the Garman-Kohlhagen PDE. State sensible boundary conditions for a vanilla call and discuss stability/convergence checks for an explicit or Crank-Nicolson finite-difference solver.

## Problem 49

[Model Validation] – Hard

You are validating an FX volatility model. List a disciplined validation suite spanning vanilla reproduction, arbitrage, parameter stability, Greeks, stress behavior, barrier benchmarks, and numerical convergence.

## Problem 50

[Final Synthesis] – Hard

`EUR/USD` spot is 1.1000. A desk is short a 3M up-and-out call with strike 1.1100 and barrier 1.1500. The 25D RR becomes 1 vol point more negative, ATM vol rises 0.7 points, and spot jumps to 1.1350. Explain qualitatively how smile, barrier proximity, delta/gamma, vanna/volga, and hedging liquidity interact. Propose a practical risk-management response.

# A Fully Worked Solutions

## Solution 1

`EUR/USD` = 1.0875 means 1 EUR costs 1.0875 USD. Under S = domestic per foreign, USD is domestic and EUR is foreign. If `EUR/USD` rises to 1.0950, EUR has strengthened versus USD. A long `EUR/USD` position profits; a short position loses.

> **Interview tip.** domestic/foreign depends on quote direction, not the trader's home country.

## Solution 2

For JPY pairs, one pip is usually 0.01. The move 149.68-149.20 = 0.48 JPY = 48 pips. In 149.683 the third decimal is a fractional pip, commonly called a pipette (0.001 JPY here).

## Solution 3

The treasurer will receive EUR and then economically convert to USD, so the risk is that EUR weakens before receipt. The firm is effectively long future `EUR/USD` exposure. To hedge, sell EUR forward / buy USD forward for the expected EUR 5m receipt date.

## Solution 4

The no-arbitrage cross is `EUR/JPY` = 1.1000*150.00 = 165.00, but market is 166.50, so EUR is expensive versus JPY on the direct cross. Start JPY 150,000,000. Convert JPY to USD: 150,000,000/150 = USD 1,000,000. Convert USD to EUR at `EUR/USD` 1.10: 1,000,000/1.10 = EUR 909,090.91. Sell EUR for JPY at 166.50: 909,090.91*166.50 = JPY 151,363,636. Profit ≈ JPY 1,363,636, about 0.91%, ignoring spreads and execution.

## Solution 5

USD/EUR = 1/1.25 = 0.8000. If `EUR/USD` rises 1% from S to 1.01S, inverse becomes 1/(1.01S) = (1/S)/1.01, a decline of about 0.9901%, approximately -1% for small moves. Formally d(1/S)/(1/S) ≈ -dS/S to first order.

## Solution 6

Borrow S0 units of domestic currency, convert at spot into 1 unit foreign, invest foreign at r_f to obtain exp(r_f T) foreign. Lock a forward to sell that amount at F. Domestic proceeds at T are F exp(r_f T). The domestic borrowing grows to S0 exp(r_d T). No-arbitrage requires F exp(r_f T)= S0 exp(r_d T), hence F=S0 exp((r_d-r_f)T).

## Solution 7

F=1.0800*exp((0.045-0.025)*0.5)=1.0800*exp(0.01)≈1.09085. Forward points = 1.09085-1.0800= 0.01085 ≈ 108.5 pips. Positive points reflect higher USD rate than EUR rate under this quote.

## Solution 8

With simple compounding, one domestic unit invested becomes 1+r_d T; one foreign unit becomes 1+r_f T. Replicating one foreign unit gives F=S(1+r_d T)/(1+r_f T). Numerically F=1.3000*(1+0.05*0.25)/(1+0.03*0.25)=1.3000*1.0125/1.0075≈1.30645.

### Solution 9

Linear time interpolation: 4M is one third of the way from 3M to 6M, so points ≈ 28 + (1/3)(61-28)= 39 pips. Outright = spot + 0.0039. Limitation: forward points are driven by discount curves and day-count conventions, so linear interpolation in raw points can distort curve shape, especially across turns or stressed markets.

### Solution 10

Spot leg: client sells EUR 10m and receives USD 10m*1.0900 = USD 10.9m. Forward leg at 1.0945: client buys EUR 10m and pays USD 10.945m at 3M. The swap exchanges near-term EUR for USD and reverses the currency exchange later; the 45-pip difference reflects carry/funding.

### Solution 11

Payoff to long USD = 1,000,000*(85.20-84.00)/85.20 = USD 14,084.51. Because fixing is above contracted NDF rate, USD strengthened versus INR, so a long-USD position receives cash.

### Solution 12

Payoff = 2,000,000*(82.80-83.50)/82.80 = -USD 16,908.21. Negative means the long-USD party pays about USD 16.9k; INR strengthened relative to the NDF strike.

### Solution 13

Fixing risk is the exposure to the exact benchmark fixing print because the settlement amount is determined by that rate, not by an arbitrary tradable spot at the same instant. Basis risk arises because offshore NDFs, onshore spot/forwards, and futures can diverge due to capital controls, funding, liquidity, and fixing methodology. A hedge can therefore be directionally correct yet leave residual P&L from benchmark mismatch, timing, and basis jumps.

### Solution 14

Under the domestic risk-neutral measure, dS=(r_d-r_f)Sdt+sigma S dW. Since foreign currency behaves like an asset with continuous yield r_f, Black-Scholes applies with dividend yield q= r_f. Therefore C=S e^{-r_f T}N(d1)-K e^{-r_d T}N(d2), where d1=[ln(S/K)+(r_d-r_f+0.5 sigma^2)T]/(sigma sqrt(T)) and d2=d1-sigma sqrt(T). The domestic discount factor prices the strike payment; the foreign discount factor reflects carry on the underlying FX asset.

### Solution 15

Compute sigma*sqrt(T)=0.12*sqrt(0.5)=0.08485. ln(1.10/1.12)=-0.01802. Drift term=(0.04-0.02+0.5*0.12^2)*0.5=0.0136. Thus d1=(-0.01802+0.0136)/0.08485=-0.0521; d2=-0.1369. Using N(d1)≈0.4792, N(d2)≈0.4455, C≈1.10*e^-0.01*0.4792 -1.12*e^-0.02*0.4455 ≈0.5220-0.4891=0.0329 USD per EUR notional.

### Solution 16

Parity follows by subtracting the GK put from call: C-P=S e^{-r_fT}-K e^{-r_dT}. So P=C-S e^{-0.01}+K e^{-0.03}. Numerically S e^-0.01≈1.18806; K e^-0.03≈1.14513. P≈0.075-1.18806+1.14513= 0.03207.

### Solution 17

At expiry, long call minus long put payoff is $\max(S_T-K,0)-\max(K-S_T,0)=S_T-K$. That is the payoff of a long forward struck at K, but the time-0 value is not generally zero unless K equals the market forward. Discounted parity gives $C-P=S e^{-r_fT}-K e^{-r_dT}$. For K=F, the RHS is zero and call-put is a zero-cost forward replication.

### Solution 18

Spot delta = $e^{-r_fT}N(d1)=e^{-0.02*0.5}*0.6368\approx 0.6305$. Forward delta = $N(d1)=0.6368$ under the standard unadjusted convention. Forward delta removes the foreign discounting factor.

### Solution 19

Spot delta is sensitivity to spot and commonly includes $e^{-r_fT}$; forward delta is sensitivity expressed against the forward and typically uses N(d1). Premium-adjusted delta subtracts the premium-related underlying amount because premium itself is paid in one currency and changes effective hedge units. Since market quotes often define 25D by a specific delta convention, changing convention changes the strike solving Delta(K)=0.25.

**Interview tip.** always ask which delta and premium currency convention is being used.

### Solution 20

Gamma $\approx e^{-r_fT}$ n(d1)/(S sigma sqrt(T)). Here $e^{-0.005}\approx 0.9950$ and denominator=1.10*0.10*0.5=0.055. Gamma$\approx$0.995*0.391/0.055$\approx$7.07 per unit S. Vega = S $e^{-r_fT}$ n(d1)sqrt(T)$\approx$1.10*0.995*0.391*0.5$\approx$0.214. This is per 1.00 absolute volatility; per 1 vol point divide by 100, $\approx$0.00214.

### Solution 21

ATM-spot uses K=S. ATM-forward uses K=F. Delta-neutral ATM chooses strike so call and put deltas offset under a specified convention; this may differ from both S and F. FX has two rates and market delta conventions, so 'ATM' is convention-dependent. Under ATM-forward, K=F by definition.

### Solution 22

Let c=$\sigma_{25C}$ and p=$\sigma_{25P}$. RR=c-p=-1.2 and (c+p)/2 -10.0=0.5, so c+p=21.0. Solving: 2c=19.8 -> c=9.9%; p=11.1%. Negative RR means put wing vol exceeds call wing vol.

### Solution 23

With forward delta 0.25, $d1=N^{-1}(0.25)=-0.67449$. In forward form $d1=[\ln(F/K)+0.5\sigma^2 T]/(\sigma\sqrt{T})$. Rearranging $\ln(F/K)=d1\,\sigma\sqrt{T}-0.5\sigma^2 T$. Hence $K=F\exp(-d1\,\sigma\sqrt{T}+0.5\sigma^2T)$. $\sigma\sqrt{T}=0.08485$. Exponent=0.67449*0.08485+0.0036$\approx$0.06084. K$\approx$1.10*$e^{0.06084}\approx$1.1691.

### Solution 24

A practical scheme is: convert delta quotes to strikes consistently; interpolate total variance $w(T,k)=\sigma^2 T$ across maturity; use an arbitrage-aware smile parameterization (e.g. SVI-style or constrained spline) across log-moneyness; then map back to delta if needed. Checks: call prices must decrease with strike, be convex in strike, and preserve calendar consistency of appropriately normalized option values/total variance. Also check positive density, stable wing extrapolation, and no discontinuities near ATM.

## Solution 25

For fixed maturity, call price C(K) must be non-increasing: dC/dK<=0. It must be convex: d^2C/dK^2>=0, equivalent to nonnegative risk-neutral density. Bounds include 0<=C<=S e^{-r_fT}. Across maturity, one must compare consistently normalized prices/forwards; careless comparison of raw implied vols can produce false calendar-arbitrage conclusions because forwards and discount factors change.

## Solution 26

Dupire chooses a deterministic local volatility sigma_loc(S,t) so that the model matches the entire continuum of vanilla prices. It is effectively inferred from derivatives of the option surface. Barriers depend on path behavior, not only terminal distributions. Local vol can generate unrealistic spot-vol co-movement and conditional path dynamics, so two models fitting the same vanillas may value barriers differently.

## Solution 27

A common FX Heston specification: dS=(r_d-r_f)Sdt+sqrt(v)S dW_S; dv=kappa(theta-v)dt+xi sqrt(v)dW_v; corr(dW_S,dW_v)=rho dt. kappa controls mean reversion, theta long-run variance, xi vol-of-vol, rho spot-variance correlation. r_d and r_f determine the domestic-measure drift.

## Solution 28

Negative spot-vol correlation means downward spot shocks tend to coincide with rising variance. Under S=domestic per foreign, this typically lifts implied vols on the lower-strike/put side, producing a negatively sloped skew in strike space. Exact market interpretation still depends on quote direction and which currency is considered the risk asset.

## Solution 29

In SABR intuition, alpha sets the volatility level, beta controls elasticity/backbone, rho mainly drives skew, and nu (vol-of-vol) drives smile curvature. If every maturity has four unconstrained parameters, many combinations fit sparse quotes almost equally well, causing unstable term structures and noisy Greeks. Fixing beta, smoothing parameters over maturity, and regularization usually improve robustness.

## Solution 30

One objective is weighted least squares over market instruments: sum_i w_i[modelVol_i(theta)-marketVol_i]^2 + lambda R(theta), where instruments include ATM, 25D and 10D RR/BF across maturities. Choose w_i roughly inverse bid-ask variance so liquid tight quotes matter more. R penalizes abrupt parameter jumps or implausible values. Validate out-of-sample strikes, parameter continuity, and Greek stability; a tiny calibration error with unstable parameters is not a good model.

### Solution 31

An up-and-out ceases to exist if spot hits an upper barrier; up-and-in activates only after that hit. Down variants use a lower barrier. For same strike, maturity, barrier and rebate conventions, vanilla = knock-in + knock-out. Thus KI = 0.042-0.027=0.015 USD per unit foreign notional.

### Solution 32

Daily monitoring misses some intraday barrier touches, so the discretely monitored knock-out is generally more valuable than the continuously monitored knock-out because it has a lower probability of being extinguished. The difference is a barrier-monitoring correction and grows when volatility is high or spot is near the barrier.

### Solution 33

Value depends strongly on distance-to-barrier, volatility, skew, rates/carry, and time. As spot approaches the barrier, hit probability becomes highly sensitive to small spot changes; delta and gamma can become very large and unstable. If payoff occurs immediately on touch, timing and discounting also matter. Hedging can be difficult because the position can disappear abruptly once the barrier is hit.

### Solution 34

A double-no-touch pays only if spot stays inside two barriers until expiry; its risk is dominated by survival probability and both tails. A TARF is a path-dependent structured product with repeated fixings and termination after an accumulated target is reached, often creating asymmetric leverage. Model risks: DNT - smile dynamics, barrier monitoring, joint tail/path behavior. TARF - path dependence, forward skew, fixing correlations, gap risk, and client-unwind behavior.

### Solution 35

Average A=(1.09+1.10+1.12+1.11)/4=1.1050. Call payoff=max(A-K,0)=max(1.1050-1.1050,0)=0. Averaging reduces effective volatility versus a single terminal fixing and therefore often lowers option value relative to a comparable vanilla.

### Solution 36

A quanto fixes the conversion rate between the foreign underlying payoff and domestic settlement, removing direct FX payoff exposure but not the statistical dependence between underlying and FX. Changing measure introduces a covariance term. In a common setup the foreign underlying's domestic-measure drift is shifted by a term proportional to -rho*sigma_X*sigma_S. The exact sign depends on whether S is domestic/foreign or its inverse and on the payoff definition.

> **Interview tip.** derive the sign from the chosen quote and numeraire; do not memorize it blindly.

### Solution 37

Since `EUR/GBP` = (`EUR/USD`)/(`GBP/USD`)=1.0800/1.2600=0.85714 GBP per EUR. Market 0.8500 is about 0.83% lower, which is not exactly consistent and could imply arbitrage if executable bid-ask quotes allow the cycle after transaction costs.

### Solution 38

Each standalone gamma assumes its own pair moves independently. A common USD factor drives both `EUR/USD` and `GBP/USD`; joint moves create covariance terms in portfolio P&L, schematically dP≈...+0.5 Gamma_E dS_E^2+0.5 Gamma_G dS_G^2+CrossGamma dS_E dS_G. Correlation changes the distribution of combined moves and can materially alter stress losses and hedging effectiveness.

### Solution 39

In collateralized pricing, the collateral account defines the effective funding/discounting economics. If cash flows are collateralized in currency c, the relevant discount factors and forward relationships are linked to c-collateralized curves, not simply unsecured domestic and foreign rates. Changing CSA currency can therefore change discount factors, implied forwards, and option present values even with the same spot.

### Solution 40

F=1.1000*(0.9850/0.9700)=1.1000*1.0154639≈1.11701. The forward premium versus spot is about 0.01701 = 170.1 pips.

### Solution 41

Cross-currency basis is the spread required to reconcile actual market funding/collateral conditions with idealized CIP. It is observed through cross-currency swaps and FX swaps/forwards relative to OIS curves. Persistent basis reflects balance-sheet costs, liquidity, demand for specific currencies, regulatory constraints, and collateral conventions.

### Solution 42

Setup 1: F1=1.08*(0.975/0.955)=1.10262. Setup 2: F2=1.08*(0.972/0.960)=1.09350. Difference= 0.00912, about 91.2 pips. This illustrates that changing collateral/discount assumptions can materially shift forwards.

### Solution 43

Delta P&L: a 0.5% spot rise means dS≈1.10*0.005=0.0055 USD/EUR. Delta +EUR 4m gives ≈4m* 0.0055=USD 22,000. Gamma term using the book's stated unit convention: 0.5*0.8m*(0.5)^2≈USD 100,000 if the quoted gamma P&L is per squared 1% move; this unit convention must be confirmed. Vega P&L=25k*0.4=USD 10,000. Theta=-USD 8,000. Estimated total ≈USD 124,000 under that interpretation.

> **Interview tip.** Greek units are desk-specific; clarify whether gamma is curvature in price units or already normalized to a 1% move.

### Solution 44

For a delta-hedged long option, positive gamma means the hedge naturally buys low and sells high as spot oscillates; theta is typically negative and represents the cost of owning convexity. In the idealized continuous model, the hedged P&L is approximately 0.5*Gamma*S^2*(realized variance - implied variance)dt. Thus realized volatility above implied tends to benefit long gamma, before transaction costs, jumps, smile changes, and discrete hedging effects.

### Solution 45

Vanna is sensitivity of vega to spot, equivalently sensitivity of delta to volatility. Volga (vomma) is sensitivity of vega to volatility. Exotics can have modest first-order vega but large higher-order exposure because smile moves when spot moves and because their payoff depends strongly on where volatility is redistributed across strikes. This is especially important for barriers near the smile wings.

### Solution 46

The vanna-volga method starts with a base Black-Scholes/GK price. Construct weights in three liquid benchmark vanillas so the benchmark portfolio matches the exotic's vega, vanna, and volga. Observe each benchmark's market price minus base-model price. Apply the same weights to these smile premia and add the weighted correction to the exotic's base price. It is a practical approximation, not a fully dynamic arbitrage-free model.

### Solution 47

Simulate under domestic measure: S_{t+dt}=S_t*exp[(r_d-r_f-0.5 sigma^2)dt+sigma sqrt(dt) Z]. For each path compute arithmetic average A and payoff max(A-K,0), then discount by exp(-r_d T). Estimator is the sample mean of discounted payoffs; standard error = sample std/sqrt(N). Variance reduction: antithetic variates and a control variate using a geometric-average Asian option or terminal vanilla with known/cheap value.

### Solution 48

GK PDE: V_t + 0.5 sigma^2 S^2 V_SS +(r_d-r_f)S V_S - r_d V = 0. Vanilla call terminal condition V(S,T)=max(S-K,0). Boundary: V(0,t)=0; for large S, V≈S e^{-r_f tau}-K e^{-r_d tau}. Explicit schemes require CFL-like step restrictions; Crank-Nicolson is unconditionally stable in the linear sense but can oscillate near kinks unless damped. Check grid refinement, time-step refinement, domain truncation, monotonicity, and agreement with analytic GK prices.

### Solution 49

A disciplined validation suite includes: (1) reproduce calibration vanillas within bid-ask; (2) test out-of-sample strikes/maturities; (3) enforce/diagnose static arbitrage; (4) inspect parameter term structures and day-to-day stability; (5) finite-difference Greeks versus analytic/adjoint where available; (6) stress spot, vol, rates, correlation and basis; (7) compare barriers against alternative models/benchmarks; (8) check MC confidence intervals and PDE grid convergence; (9) assess hedging P&L on historical scenarios; (10) document limitations and model-change triggers.

### Solution 50

The short up-and-out call is close to its 1.1500 barrier after spot jumps to 1.1350, so barrier hit probability rises sharply. Higher ATM vol increases both vanilla optionality and touch probability; for a knock-out, those effects compete, and barrier proximity often dominates. A more negative RR means downside puts richen versus calls, changing local skew around the relevant strike/barrier and therefore the conditional barrier dynamics. Delta/gamma can become unstable near the barrier, while vanna/volga matter because spot and vol moved together. A practical response is to re-mark under the current smile, run barrier-distance and gap stresses, reduce oversized delta/gamma with spot/forwards and liquid vanillas, hedge smile with 25D/10D instruments where possible, set tighter intraday limits, and assess liquidity if the barrier is touched. Avoid relying on a single local-vol or Black-Scholes hedge; compare model variants and include discrete-monitoring/gap risk.

## Final Review Checklist

Before an FX-derivatives interview, make sure you can:

- switch quote direction without losing the domestic/foreign convention;
- derive CIP and Garman–Kohlhagen rather than merely quote formulas;
- explain spot, forward, and premium-adjusted delta conventions;
- reconstruct 25D wing volatilities from ATM/RR/BF;
- distinguish terminal-distribution fit from path-dependent exotic pricing;
- discuss collateral, cross-currency basis, and model risk;
- perform a Greek P&L explain while checking units and sign conventions.